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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

R V Vidyanikethan Post
Mysuru Road Bengaluru - 560 059

V Semester B.E. Regular / Supplementary Examination February/ March 2026

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Machine Learning Operations

Time : 3 Hours

Maximum Marks : 100

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Define model versioning and explain its importance in the machine learning lifecycle.	02	1	2
1.2	Identify the challenge that arises when there is a communication gap between data science teams and software engineering teams.	02	1	2
1.3	How does governance support Responsible AI in MLOps?	02	4	2
1.4	A company wants to deploy a new ML model without interrupting users. Which MLOps practice should be used?	01	3	2
1.5	An e-commerce company deploys a recommendation model that performs well during testing. After deployment, the model works correctly but starts responding slowly during peak shopping hours, affecting user experience. a) Identify the issue faced by the deployed model. b) Which MLOps practice or feature helps address this issue?	01	4	2
1.6	Name two types of adversarial threats affecting ML models	02	2	1
1.7	Identify two risks associated with deploying ML models without proper validation.	02	1	2
1.8	Give reason for the need of a centralized version control system essential in a continuous integration pipeline?	02	4	2
1.9	How does domain classifier work as a data drift detection technique.	02	2	4
1.10	Give any two features of monitoring using Azure ML	02	3	2
1.11	What are the two operations specific to ML that a cloud provider need to monitor?	02	2	2

Part B

Question No	Question	M	CO	BT
2a	A retail company, Trend Mart, aims to improve inventory management using a machine learning demand forecasting model. The forecasting depends on seasonal sales trends, promotions, supply chain delays, and market fluctuations. TrendMart struggled with pipeline maintenance due to frequent retraining needs and integration issues with the existing ERP system. Draw a neat diagram of the MLOps pipeline and explain how you design, integrate, and automate an end-to-end MLOps system for accurate and scalable demand forecasting.	10	3	4
2b	Explain the role of collaboration between Software Engineers, DevOps Engineers, Model Risk Managers and Machine Learning Architect in ensuring successful MLOps implementation. Support your answer with an example each.	06	2	2
3a	Discuss the importance of feature selection and its impact on model performance.	06	3	2
3b	Explain the complete lifecycle of Machine Learning Operations (MLOps), including model development, deployment, monitoring, iteration, and governance. Provide examples for each stage.	10	3	2
	OR			
4a	Compare different model evaluation metrics used in ML.	06	4	3
4b	A hospital is building a disease prediction ML system using patient health records. The model must comply with strict regulations, auditability, reproducibility, and ethical use of data. Design a comprehensive MLOps governance framework for the hospital covering:Data governance and compliance,Model auditability and reproducibility,Ethical considerations,Version control for models and datasets and Monitoring for biased predictions.	10	3	3
5a	Explain how reproducibility and auditability support governance in MLOps systems.	06	2	2
5b	Explain the concept of model risk evaluation and discuss the major origins of ML model risk.	06	2	2
5c	Explain the challenges involved in adapting machine learning models from development to production environments.	04	2	2
	OR			
6a	Analyze the challenges of deploying ML models in heterogeneous production environments and explain how MLOps practices address these challenges.	08	3	4
6b	Explain machine learning security with emphasis on adversarial and data poisoning attacks. How can MLOps mitigate these risks?	08	3	3
7a	Discuss the role of blue–green deployment in upgrading models for real-time machine learning systems, focusing on how maintaining two parallel environments helps avoid service interruptions and limit the impact of faulty releases. Give an example for the same	08	2	1

7b	Explain how automated testing pipelines such as Jenkins-based CI/CD reduce deployment failures in ML projects. Give example.	08	2	3
	OR			
8a	Analyze the importance of a well-structured testing pipeline in MLOps. How do base-case datasets, edge-case datasets, and recent production datasets together improve the reliability and diagnosability of ML models during deployment?	08	4	4
8b	A retail company uses a machine learning model to predict weekly sales across its stores. Over time, customer behavior changes due to seasonal trends, new product launches, and regional festivals. Recently, the model's accuracy has started dropping because the input data patterns are no longer consistent with the original training data. Apply your understanding of model maintenance to suggest how often the ML model should be retrained and discuss the factors affecting model retainment?	08	2	3
9a	An ML model deployed on a cloud platform starts producing inconsistent predictions, although the application is running without failures. Explain and analyze how Python logging help to diagnose this issue.	06	3	4
9b	Describe observability in Cloud MLOps with reference to logging, monitoring, and drift detection in AWS SageMaker.	10	3	1
	OR			
10a	With a neat diagram, discuss Amazon cloud ML integration with ML Models.	10	3	2
10b	Discuss the common model monitoring metrics that can be set for various applications	06	4	2